	Target engagement	Dose	Efficacy	Safety
#6	**Alzheimer disease** HRH3 antagonists (3)	**2014** Pig PET and in-vitro H3 affinity of GSK239512 (vs human [11C]GSK189254 PET)	**2014** Rat five-trial inhibitory… Target exposure reached; ADAS-Cog failed	**2016** Rodent procognitive battery… Three rodent successes, three AD failures	**2016** Rat safety pharmacology an… Sponsor confirms dose-dependent sleep disturbance
	Alzheimer disease HTR6 antagonists (3)	**2023** Non-human primate PET 5-HT6 receptor occupancy… Rat under-predicted human brai…	**2022** Rat cognition battery (object… Rat efficacious exposure reached; trial failed	**2022** Rat cognition battery (object… Rat efficacy did not translate; Phase 2 null	**2014** IND-enabling toxicology (species and findings unpublished; no NDA, so no FDA review)
	Alzheimer disease anti-APP antibodies (3)	**2021** tg-ArcSwe mice, mAb158 Mice: no plaque fall; patients: PET fell	**2014** hAPP/PS1 mice, crenezumab 15 mg/kg failed; higher doses failed too	**2014** hAPP/PS1 mice, crenezumab Crenezumab missed both primaries	**2017** Aged APP mice given 2H6 (ponezumab parent) Excess microbleeds missed
#7	**asthma** PTGDR2 antagonists (9)	**2016** Human eosinophils and Th2 cells in vitro	**2020** Eosinophil DP2 potency Target reached; exacerbations unchanged	**2014** Allergen challenge Challenge said yes; FEV1 trials missed	**2012** Off-target selectivity panels; animal toxicology unpublished
	asthma ADRB2 agonists (4)	**2017** Tanabe-era in vitro receptor-binding kinetics of TA-2005 (carmoterol):…	**2014** AstraZeneca guinea-pig histamine bronchoconstriction,… 2 ug once daily missed 24-h…	**2014** AstraZeneca guinea-pig histamine bronchoconstriction,… 24-h trough FEV1 primary missed	**2017** Guinea-pig isolated right atrium, left atrium and papillary muscle chronotropic/inotropic assa…
	asthma NR3C1 agonists (4)	**2017** Rat allergic airway inflammation (lung oedema; thymic involution as systemic index)	**2020** Rat allergic airway inflammation (lung oedema; thymic… Top dose missed week-12 FEV1	**2018** Allergen-induced late asthmatic response in mild atopic… Half-ICS LAR effect; 1 mg trial failed	**2020** Rat allergic airway inflammation (lung oedema; thymic involution as systemic index)
	asthma anti-IL13 antibodies (4)	**2013** IL-13-stimulated bronchial epithelial cells and patient brushin… Airway eosinophils did not fall	**2018** Integrated population PK/PD dose-exposure-FEV1 model of… Model dose reached; phase 3 failed	**2018** IL-13-humanised… Phase 3 exacerbation and OCS primaries missed	**2022** Cynomolgus repeat-dose toxicology (tralokinumab, lebrikizumab)
#8	**breast cancer** CDK4/6 inhibitors (6)	**2016** Human tumour xenografts in mice (phospho-Rb Ser780)	**2017** Abemaciclib xenograft PK/PD (pRb to growth) Brain exposure OK, no response	**2014** Human breast cancer cell-line panel (47 lines) Brain-metastasis trial missed	**2017** Dog cardiovascular telemetry (QTc) with hERG assays Missed liver injury and clots
	breast cancer ESR1 degrader (5)	**2026** ESR1-mutant models (MCF7-Y537S, ST941/HI Y537S PDX, mutant-ER PDX)	**2026** ESR1-mutant models (MCF7-Y537S, ST941/HI Y537S… Target reached; amcenestrant…	**2025** ESR1-mutant models (MCF7-Y537S, ST941/HI Y537S… Amcenestrant fails PFS vs…	**2026** Rat and dog repeat-dose toxicology (oral ER degraders) Clean hERG and dog telemetry,…
	breast cancer HSP90AA1/HSP90AB1 inhibitors (5)	**2014** HER2-overexpressing SKOV-3 xenograft with 89Zr-trastuzumab PET readout of…	**2013** HER2-positive breast cancer xenografts (BT-474, JIMT-1) with HER2, p-AKT and HSP70 readouts	**2013** HER2-positive breast cance… HER2+ trial: 0 of 26 responses, stopped	**2013** Mouse xenograft studies with cardiac and liver readouts… Keratitis and heart DLTs…
	breast cancer PI3K/MTOR inhibitors (5)	**2018** Novartis BEZ235 xenografts with ex-vivo tumour PI3K/Akt pathway readout	**2015** Exelixis XL765 xenograft… MTD reached, phase II failed, abandoned	**2015** Exelixis XL765 xenograft panel with dose-dependent… No responses, PFS6 8% in phase II	**2016** Genentech GDC-0980 PI3K-pathway-activated xenografts… Hyperglycaemia DLT; model silent
	breast cancer anti-IGF1R antibodies (4)	**2011** ER+ MCF-7 breast xenografts in nude mice (IMC-A12, h7C10) GFS fell in fewer than 40%…	**2015** ER+ MCF-7 breast xenogra… Target trough reached; drug failed; abandoned	**2017** Tamoxifen-resistant MCF-7L cells and tamoxifen-treated… Adding dalotuzumab to mTOR…	**2020** Cynomolgus monkey… Monkeys missed hyperglycaemia (10% in TED)
	breast cancer PARP1/2 inhibitors (3)	**2021** Pamiparib's own BRCA1-mutant MDA-MB-436 TNBC xenograft with tumour PAR…	**2021** Pamiparib's own BRCA1-mutant MDA-MB-436 TNB… Target reached, potentiation failed	**2022** Pamiparib's own BRCA1-mutant MDA-MB-436 TNB… Veliparib adds nothing to…	**2020** 2X-121 (E7449) BRCA-deficient xenografts:… Anaemia, neutropenia unforese…
	breast cancer PI3K (pan) inhibitors (3)	**2014** PI3K-pathway-activated lines and xenografts (U87MG, IGROV-1; breast lines incl. MCF7, KPL-4)	**2016** PI3K-pathway-activated line… Target exposure reached, pictilisib failed	**2016** Breast cancer patient-derive… PIK3CA-mutant enrichment not seen in FERGI	**2017** PI3K-pathway-activated line… Liver toxicity unflagged by preclinical package
	breast cancer PIK3CA inhibitors (3)	**2017** PIK3CA-mutant cell-line panel (CCLE) and PIK3CA-mutant breast cancer xenografts	**2019** PIK3CA-mutant cell-line panel (CCLE) and PIK3CA-mutant breast cancer xenografts	**2020** Patient-derived xenograft of the index patient's PTEN-null, BYL719-refractory lung metastasis…	**2024** Sponsor toxicology and secondary pharmacology (mouse insulin/glucose tolerance, rat and…
	breast cancer pan-HER inhibitors (3)	**2014** ER+/HER2-negative tamoxifen-resistant MCF7 and T47D lines and MCF7/TamRes xenograft…	**2017** HER2-amplified breast lines and xenografts (BT-474, SK-BR-3)	**2019** HER2-amplified breast lines and xenografts (BT-474, SK-BR-3)	**2017** HER2-amplified breast lines… Diarrhoea DLT despite stated animal window
#9	**osteoarthritis** MAPK14 inhibitors (3)	**2012** Rat iodoacetate joint degeneration + carrageenan thermal hyperalgesia (SB-203580, VX-745)	**2009** Rat and cynomolgus endotoxin-induced TNF-alpha PD model (PH-797804)	**2012** Rodent rheumatoid-arthritis models with ex-vivo LPS whole-blood cytokine PD…	**2014** Rat repeat-dose intra-articular toxicology (FX005)
	osteoarthritis NTRK1 inhibitors (3)	**2019** Rat MIA and MNX knee OA (AR786, sibling TrkA inhibitor)	**2019** Rat MIA and MNX knee OA (AR786, sibling TrkA inhibitor)	**2019** Rat PGPS knee arthritis flare (GZ389988, single intra-articular… Knockout says TrkA block enoug…	**2021** Rat 4- and 13-week oral neurotoxicity (ASP7962) Rat silent; two-thirds got joint…
#10	**colorectal cancer** KDR (VEGFR) inhibitors (3)	**2014** Mouse lung VEGF-induced pVEGFR2 (fruquintinib)	**2014** Mouse lung VEGF-induced pVEGFR2 (fruquintinib)	**2017** Human colorectal patient-derived xenografts (PDX) in mice	**2023** Dog cardiovascular safety… Dogs silent; 14% grade 3-4 hypertension
	colorectal cancer VEGFR multikinase inhibitors (3)	**2009** VEGF-induced uterine edema in mice (vascular-permeability PD model, ABT-869/linifanib)	**2009** Human tumour cell-line xenografts in mice incl. colon carcinoma (ABT-869/linifanib)	**2017** Human tumour cell-line xenografts in mice incl. colon carcinoma (ABT-869/linifanib)	**2017** Conscious telemetered rats… Hypertension in patients as rats predicted
	colorectal cancer anti-EGFR antibodies (3)	**2018** EGFR-expressing cancer cell lines and A431 xenografts (Sym004)	**2018** A431NS xenograft PK and… Dose reached; OS failed; program dropped	**2018** EGFR-expressing cell lines… Phase 2 OS failed; ECD biomarker failed	**2015** Cynomolgus repeat-dose toxicology with cetuximab comparator (Sym004)
#11	**chronic hepatitis B virus infection** HBV RNA antisense / siRNA (3)	**2022** Chronically HBV-infected chimpanzees	**2018** HBV-infected human liver-chimeric mice (PXB uPA/SCID with human hepatocytes)	**2023** HBV-genome mice (pHBV transient/transgenic, later AAV-HBV)	**2023** HBV-infected human liver-chimeric mice (PXB uPA/SCID with human hepatocytes)
	chronic hepatitis B virus infection HBsAg release (NAP) inhibitors (3)	**2020** DHBV-infected Pekin ducks Human hepatocytes miss the…	**2020** CD-1 mouse and cynomolgus PK/toxicology (uninfected)	**2016** DHBV-infected Pekin ducks	**2020** CD-1 mouse and cynomolgus PK/toxicology (uninfected)
#12	**hepatocellular carcinoma** VEGFR multikinase inhibitors (4)	**2015** Human HCC cell-line and patient-derived xenografts in immunodeficient mice	**2018** Population PK / exposure-response model of lenvatinib in HCC (phase 1 pool +…	**2020** Mouse HCC with diverse genetic drivers (hydrodynamic transposon delivery) screened for…	**2015** GLP repeat-dose toxicology in rat plus dog or monkey
	hepatocellular carcinoma anti-PDCD1 antibodies (4)	**2025** In-vitro PD-1 receptor-occupancy titration of penpulimab on human…	**2025** Cynomolgus monkey 6- and 13-week repeat-dose toxicology of penpulimab (4, 15, 60 mg/kg every…	**2023** Syngeneic orthotopic mouse HCC (HCA-1 in C3H, RIL-175 in C57BL/6; grafted and genetically…	**2022** PD-1 knockout mice Monkeys silent; patients had irAEs
	hepatocellular carcinoma KDR (VEGFR) inhibitors (3)	**2020** Human liver-cancer cell-line xenografts in athymic rats and nude mice (KRN951 rat panel incl.…	**2020** Human liver-cancer cell-line xenografts in athymic rats and nude mice (KRN951 rat panel incl.…	**2021** Syngeneic murine HCC allograft in immunocompetent C57BL/6 mice (apatinib…	**2021** GLP repeat-dose toxicology and cardiovascular safety pharmacology in rodent plus dog o…
	hepatocellular carcinoma anti-CD274 antibodies (3)	**2021** Primary human T-cell activation assay (proliferation, IFN-gamma) with durvalumab; file…	**2022** Cynomolgus monkey 4-week and 13-week GLP repeat-dose toxicology of durvalumab (up to…	**2022** Syngeneic CT26 mouse colorectal carcinoma dosed with a rat anti-mouse PD-L1 surrogate…	**2025** Cynomolgus monkey 4-week and 13-week GLP repeat-dose toxicology of durvalumab (up to…
#13	**schizophrenia** PDE10A inhibitors (5)	**2019** Rat and monkey ketamine-challenge phMRI/EEG reversal	**2019** Rodent occupancy dosing Predicted-adequate exposure, no effect	**2017** Ketamine challenge Challenge passed; phase 2 PANSS missed	**2017** Rat catalepsy, prolactin and glucose vs D2 antagonists EPS in patients at clinical dose
	schizophrenia SLC6A9 inhibitors (4)	**2018** Rat CSF/striatal glycine by microdialysis Human ketamine WM readout di…	**2025** Rat CSF/striatal glycine by microdialysis 10 mg hit PD target; failed	**2023** Human ketamine-challenge working-memory fMRI in healthy… Three phase 3s flat on MCCB	**2026** Rodent haematology (haemoglobin) in pharmacology and toxicology
	schizophrenia CHRNA7 agonists (3)	**2024** Pig [11C]NS14492 PET alpha7 occupancy	**2015** Rodent schizophrenia screens: th(tk-)/th(tk-) mouse PPI… Above rat-effective dose, no effect	**2016** Rodent schizophrenia screens: th(tk-)/th(tk-) mouse PPI… ABT-126 phase 2b missed MCCB	**2012** Rodent schizophrenia screens: th(tk-)/th(tk-) mouse PPI and social behaviour, rat…
#14	**gastric cancer** PARP1/2 inhibitors (3)	**2023** Gastroesophageal adenocarcinoma PDX platform and PDX-derived primary cells, BRCA2…	**2023** Gastroesophageal adenocarcinoma PDX platform and PDX-derived primary cells, BRCA2…	**2026** Gastroesophageal adenocarcinoma PDX platform an… Hypoxia-sensitisation bar misse…	**2017** Rat and dog general toxicology, up to 26 weeks (olaparib)
	gastric cancer anti-ERBB2 antibodies (3)	**2017** Trastuzumab/pertuzumab-derived anti-HER2 bispecific on Alphamab's heterodimeric (CRIB) Fc, cancer-c…	**2020** Cynomolgus monkey single and repeat-dose toxicology and… Target reached, JACOB failed	**2026** Trastuzumab/pertuzumab-derived anti-HER2 bispecific on Alphamab'… JACOB missed overall survival	**2020** Cynomolgus monkey single and repeat-dose toxicology and toxicokinetics of MGAH22…
#15	**hepatitis C** HCV NS3/4A protease inhibitors (13)	**2013** Chronically HCV-infected chimpanzee (genotype 1a/1b)	**2009** Replicon EC99 vs rat plasma and liver exposure	**2016** Human hepatocyte chimeric mouse (uPA/SCID) infected with patient HCV genotype 1b	**2016** Rodent and dog repeat-dose toxicology